Chafon Hardware Bridge Manual

Digitex School Management System

Version: Laravel 11 | **Document date:** June 22, 2026

E-Digitex SMS — Multi-Institution Education Platform

Chafon CF661 RFID Bridge — Setup, Configure, Test & Run

Digitex School Management System — Hardware Attendance

This guide explains how to connect a **Chafon CF661** UHF RFID reader to Digitex using the Python bridge script (`chafon-script.py`) on a local PC or Raspberry Pi.

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1. How authentication works (important)

Digitex hardware scans do **not** use per-school API keys such as `dgtx_live_…`.

Layer	What you configure	Who sets it
Server	`HARDWARE_SECRET` in Laravel `.env`	Platform / Super Admin (once per server)
Each gate PC	Same `HARDWARE_SECRET` + that school's `HARDWARE_INSTITUTION_ID`	School IT or installer

- **One shared secret is enough for all schools** on the same Digitex installation.
- **Each school** must set a **different** `HARDWARE_INSTITUTION_ID` on its own bridge so scans are routed to the correct school.
- Schools do **not** enter an API key in the web UI for hardware; they only configure the bridge environment on the gate computer.

Optional server setting:

```
HARDWARE_ALLOWED_INSTITUTION_IDS=3,5,12
```

When set, only those institution IDs are accepted via the `X-Institution-Id` header.

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2. Prerequisites

On the Digitex server

- PHP/Laravel app deployed and reachable over HTTPS.
- In `.env`:

```
HARDWARE_SECRET=generate-a-long-random-string-here
# Optional — comma-separated school IDs this deployment may serve:
# HARDWARE_ALLOWED_INSTITUTION_IDS=3,5
```

- Run `php artisan config:clear` after changing `.env`.

On the gate PC / Raspberry Pi

- Python 3.8+
- Package: `requests` (`pip install requests`)
- Same LAN as the Chafon reader
- Copy from the SMS repo:

\- `chafon-script.py`

\- `chafon.env.example` → rename/copy to `chafon.env`

Find your school Institution ID

In Digitex web admin: note the school context, or ask Super Admin. It is the numeric `institution_id` (e.g. `3` for Complexe Scolaire Integrale).

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3. Configure the Chafon CF661 reader

- Connect the reader to power and Ethernet.
- On a Windows PC, open **Chafon UHF RFID Demo Software**.
- **TCP/IP** tab → connect (default reader IP often `192.168.1.190`, port `6000` or `27011`).
- **Work Mode** → set to **Active Mode** (auto push on tag read).
- **Destination IP** → IP of the PC running the bridge (e.g. `192.168.1.100`).

- **Destination Port** → `5000` (must match `CHAFON_LISTEN_PORT`).
- **Save** settings to the reader.

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4. Configure the bridge (`chafon.env`)

Edit `chafon.env` on the gate machine:

```
HARDWARE_SECRET=same-value-as-server-HARDWARE_SECRET
HARDWARE_API_URL=https://your-domain.com/api/v1/hardware/attendance/scan
HARDWARE_INSTITUTION_ID=3
HARDWARE_DEVICE_ID=CHAFON_MAIN_GATE_01
HARDWARE_PURPOSE=attendance
CHAFON_LISTEN_HOST=0.0.0.0
CHAFON_LISTEN_PORT=5000
```

Variable	Required	Description
`HARDWARE_SECRET`	Yes	Must match server `.env`
`HARDWARE_API_URL`	Yes	Full URL to scan endpoint
`HARDWARE_INSTITUTION_ID`	Yes*	School ID (*optional only if server sets `HARDWARE_ALLOWED_INSTITUTION_IDS` with a single ID)
`HARDWARE_DEVICE_ID`	No	Label in logs (default `CHAFON_MAIN_GATE_01`)
`HARDWARE_PURPOSE`	No	`attendance`, `fee_check`, `pickup`, `report_card`, `identity_check`
`CHAFON_LISTEN_PORT`	No	TCP listen port (default `5000`)

Multi-school example

School	Gate PC	`HARDWARE_INSTITUTION_ID`	`HARDWARE_SECRET`
School A	PC at gate A	`3`	same platform secret
School B	PC at gate B	`5`	same platform secret

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5. Run the bridge

Windows (PowerShell)

```
cd D:\path\to\sms
pip install requests
Get-Content chafon.env | ForEach-Object {
  if ($_ -match '^([^#=]+)=(.*)$') {
    [Environment]::SetEnvironmentVariable($matches[1].Trim(), $matches[2].Trim(),
'Process')
  }
}
python chafon-script.py
```

Linux / Raspberry Pi

```
cd /opt/digitex-bridge
pip3 install requests
set -a && source chafon.env && set +a
python3 chafon-script.py
```

Run as a background service (Linux, systemd example)

Create `/etc/systemd/system/digitex-chafon.service`:

```
[Unit]
Description=Digitex Chafon RFID Bridge
After=network.target

[Service]
Type=simple
WorkingDirectory=/opt/digitex-bridge
EnvironmentFile=/opt/digitex-bridge/chafon.env
ExecStart=/usr/bin/python3 /opt/digitex-bridge/chafon-script.py
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable digitex-chafon
sudo systemctl start digitex-chafon
sudo systemctl status digitex-chafon
```

Expected console output:

```
Bridge target: https://your-domain.com/api/v1/hardware/attendance/scan
Purpose: attendance | Device: CHAFON_MAIN_GATE_01
Institution: 3
Chafon bridge listening on 0.0.0.0:5000
```

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6. Test the connection

Step A — Test server secret (curl)

From any machine with network access to the server:

```
curl -X POST "https://your-domain.com/api/v1/hardware/attendance/scan" \
  -H "Content-Type: application/json" \
  -H "X-Hardware-Secret: YOUR_HARDWARE_SECRET" \
  -H "X-Institution-Id: 3" \
  -d
"{\"uid\":\"TEST123\",\"method\":\"rfid\",\"device_id\":\"TEST\",\"purpose\":\"attendan
ce\"}"
```

Response	Meaning
`503 Hardware API is disabled`	`HARDWARE_SECRET` not set on server
`401 Unauthorized Hardware Device`	Secret mismatch
`400` / student not found	Secret OK — use a real student RFID/admission UID
`200` with check-in message	Full success

Step B — Test from Digitex web

- **Configuration → Test Notifications** (optional SMS path — separate from hardware).
- Register a student's **RFID UID** on the student profile (`rfid_uid` or `nfc_tag_uid`).

- Wave the tag at the reader; watch the bridge console for `API Response (200)`.

Step C — Bridge logs

- `ERROR: HARDWARE_SECRET environment variable is not set` → fix `chafon.env`.
- `Attempt N failed: Connection refused` → wrong `HARDWARE_API_URL` or firewall.
- `401` in API response → secret does not match server.

Server-side logs: `storage/logs/laravel.log` (search for `Unknown UID` or hardware errors).

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7. Student / staff UID setup

The scan API matches (case-normalized):

- Students: `rfid_uid`, `nfc_tag_uid`, `qr_code_token`, `admission_number`
- Staff: `nfc_uid`, `rfid_uid`, `employee_id`

Enter the tag's EPC/UID on the student or staff record before expecting attendance to record.

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8. Troubleshooting

Problem	Fix
No data in bridge console	Check reader Dest IP/port; firewall on gate PC for port 5000
503 from API	Set `HARDWARE_SECRET` on server
401 from API	Align `HARDWARE_SECRET` on bridge and server
403 Institution not authorized	Add school ID to `HARDWARE_ALLOWED_INSTITUTION_IDS` or fix `HARDWARE_INSTITUTION_ID`
400 X-Institution-Id required	Set `HARDWARE_INSTITUTION_ID` in bridge env
Unknown student	Register UID on student profile

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9. Security notes

- Never commit `chafon.env` or real secrets to Git.
- Use HTTPS for `HARDWARE_API_URL` in production.
- Rotate `HARDWARE_SECRET` if compromised; update all gate PCs the same day.
- One platform secret is normal; isolation between schools is by **Institution ID** and database scoping.

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10. Quick reference

Item	Value
Script	`chafon-script.py`
Env template	`chafon.env.example`
API endpoint	`POST /api/v1/hardware/attendance/scan`
Auth header	`X-Hardware-Secret`
School header	`X-Institution-Id`
Default listen	`0.0.0.0:5000`

Regenerate this PDF: `php artisan docs:generate-pdf` (includes Chafon Hardware Bridge Manual).
